

Examiner reports

Q1.

Nearly all students made good progress with part (a), with over half scoring full marks. Nearly all students successfully found the maximum possible friction and made a comparison with the applied horizontal force. The most frequently seen error was when students wrote a statement that was incorrect, often along the lines of “friction is greater than 150N so there is equilibrium”. This showed a misunderstanding of the friction model and could not be overlooked.

Part (b) was also well answered, but a lack of detail often meant that students had not fully justified their answer. Several impressive and detailed answers were seen, where students explained clearly that the horizontal component of the applied force was greater than the maximum friction so that the crate would not remain stationary.

Resolving of forces was generally successful.

Q9.

Part (a) was often done very well by students, with a reasonable number of students gaining all 11 marks. There were a few issues with the force diagram, with missing or additional forces. Many students seemed to have been helped by the answer that was given for the normal reaction and this in turn enabled many to find the friction correctly. Finding the acceleration was also done well, although there were a few sign errors and some students simply used 150 instead of $150 \cos 20^\circ$. Part (b) was much more demanding and there were far fewer correct responses. The most common error was not to take account of the changed angle when finding the normal reaction and to simply continue to use the value of 243 from part (a).

In part (c) there were some good responses, but a fair number of students incorrectly stated that the tension would be greater for a greater constant speed.

Q10.

This question was also done well by the majority of candidates. While most candidates were able to produce the printed answer for part (b), there were some issues with the force diagram and finding the tension in part (c). The two main problems with the force diagram were including extra forces, for example friction or simply poor diagrams without arrows or labels. In part (c) there were some confused responses and some candidates tried to resolve the normal reaction from part (b).

Q11.

This question was found to be quite difficult by many of the candidates. In part (a) there were a number of incorrect force diagrams, including errors such as including a force acting directly down the slope. In parts (b) and (c) there were a lot of confused answers, mainly due to incorrect attempts at resolving. For example, in part (b), one error was to resolve the normal reaction into components instead of the weight. However, many candidates gained full marks for part (d), where follow-through marks were awarded when the candidates' values from parts (b) and (c) were used correctly in the friction law to find a coefficient of friction.

Q12.

This question was also done well by many candidates. There were relatively few errors on the force diagrams. In part (b)(ii), some candidates made errors, the most common being to multiply the 30 by 0.5 or subtract the 30 from 49. Part (c) caused difficulties for some candidates. The most common errors here were to attempt to include the weight or to omit the friction force.

Q13.

Parts (a) and (c) were done well, with almost all of the candidates gaining full marks on these parts. Part (b) was a little more demanding. There were a number of correct solutions, but the two most common errors were to subtract 400 from 660 to obtain 260 N or to include the weight of the motorcycle and rider in some way.

Q14.

The candidates found part (a) of this question relatively straightforward, but part (b) was found to be very demanding, and very few correct solutions were seen. Candidates lost marks in part (a) mainly due to errors in the resolving: for example using $\sin 30^\circ$ instead of $\cos 30^\circ$ in part (a)(ii). A few candidates lost one mark because they gave their final answer for the coefficient of friction as 0.31 rather than 0.313.

In part (b), very few candidates realised that they needed to express the normal reaction in terms of the tension. This error resulted in solutions that were flawed. Most of the candidates who did express the normal reaction in terms of the tension went on to gain full marks for this part of the question, although a few did make arithmetic or algebraic errors.

Q15.

This question caused more difficulties. There were a large number of poor force diagrams. The most common errors were to include a reaction force and to show the two tensions as equal, by using the same letter to indicate the magnitude of each tension.

In part (b), there were candidates who did not show enough working to obtain full marks. Reasons for this included not showing a trigonometric term such as $\sin 30^\circ$ or $\cos 60^\circ$ and not showing how the value of 78.4 was obtained from the candidate's initial equation. A few candidates tried to resolve the weight instead of the tension.

The candidates did find part (c) easier, with some candidates only obtaining marks on this part of the question, using the printed result in part (b).

Q16.

The candidates made good attempts at this question. Some found it difficult to get going in part (a). Most of the candidates who had difficulties had tried to consider particle *A* instead of particle *B*. Generally the candidates made good attempts at parts (b) and (c). In part (d), the main problems were for candidates who found forming an equation of motion difficult. Often, these candidates had also found part (a) difficult. Most candidates were able to make an attempt at part (e) using their answers to parts (c) and (d).

Q17.

This question was also answered well by many candidates. A few lost a mark in part (a) by not showing that the two tensions were different, and a number of candidates used

incorrect moments in part (b), but spuriously obtained the given answer by approximating $\frac{40g}{1.9}$ to $21g$.

Q18.

This question was generally done very well, particularly parts (a) and (b). In part (c), some candidates simply assumed that the tension would be equal to the weight. Some of these candidates gave the same answer for part (d) and gained the mark there. A few candidates performed the calculation $4410 - 45$ and obtained an incorrect value.

Q19.

Part (a) was often answered well but candidates appeared to find the requests complex and supplemented their answers with several lines of unnecessary working. Part (b)(i) proved popular and successful, but part (b)(ii) was very discriminating.

Q20.

A surprising number of diagrams in part (a) failed to include all the forces. Part (b) was very successful, showing a good understanding of the friction law. Part (c) was done well by many, but others couldn't appreciate the significance of the constant speed, the necessary starting point of the explanation. Many were successful in part (d), although some confused the normal reaction with the vertical component of the tension. Part (e) was generally done well with candidates gaining credit for correct calculations using their components of the tension force.

Q21.

Diagrams in part (a) were clear and mostly correct, with occasional marks lost due to missing arrows or an additional or incorrect force; the reaction force being vertical and the weight being given as 'g' were the most frequent errors. Part (b) was mostly done correctly, but part (c) was less successful, with extra forces sometimes present in equilibrium equations. The friction law was usually applied in part (d) but marks were lost through missing or incorrect inequality signs and rounding of answers to 2 significant figures.

Q22.

Again, this was answered very well, often scoring full marks.

Q23.

There were some very good answers to this question, but a number of areas of difficulty were evident on some scripts. In part (a) a common error was to show the tension acting in an upward direction. Another error that appeared was to omit the normal reaction force. The vast majority of the candidates were able to produce the correct answer in part (b). Some candidates not find the tension directly, but found the reaction force first and then used this to find the tension. The fact that the tension was given helped some candidates, but there were a few solutions where not enough working was shown.

In part (c), one of the most common errors was to omit g from the moment equation, so that one of the terms was the product of a distance and a mass, rather than a distance

and a force. This often had an implication for the next stage when the candidates were finding the reaction force. Interestingly, some candidates took moments about the end of the rod, so that having an incorrect mass did not have an impact on their answer. Part (d) was generally done well.

Q24.

As a whole candidates seemed to find this question more demanding. While some made good attempts there were a number of candidates that showed weaknesses in resolving or that did not recognise that resolving was required. In part (b), there was a number of candidates who omitted the weight from their solutions, simply summing the vertical components of the two tensions.

Q25.

Good candidates tended to score well on this question while weaker candidates would only gain a few marks. Part (a) was often answered well, but candidates often missed the key assumption that was being made and stated assumptions that were not appropriate. In part (b) the most common error was to take the mass as 200 kg rather than 200 grams. When answering part (c), some candidates stated that the initial acceleration would be 8 ms^{-2} and some said that it would be zero. In the explanation, the quality of the expression was often poor and lacked clarity. Also some candidates did not explicitly state what would happen to the acceleration, but described what would happen to the velocity instead.

Q26.

This was an unusual question and produced many pleasing responses, showing good understanding of the law of static friction. Part (a) was generally done very well. Part (b), as expected, proved more discriminating, with a significant proportion stating 2.8 N as their answer, while some gave negative frictional forces. Part (c) produced many good solutions.

Q27.

Diagrams were mostly very good with only occasional lapses in the direction of the normal reaction force or in poor labelling of forces. Part (a) proved popular and successful for most candidates. In part (b)(i) a number failed to appreciate that the forces involved included the weight component, but many went on to be successful in part (ii).

Q28.

Many candidates were successful in this question, while others appeared to have little appreciation of the directions of forces and relevant components. Errors with signs and selection of trigonometrical function were relatively frequent.

Q29.

In part (a) diagrams were often disappointing, with common errors including an additional force in the direction of motion and the inclusion of acceleration or velocity as a force. Parts (b) and (c) were done quite well, but a significant number of candidates selected the wrong component when finding the required forces and some failed to give their final answer to the required degree of accuracy stated in the rubric.

Q30.

A surprising number of candidates failed to score both marks for the forces diagram in part (a), with the most common errors being the omission of the friction force or stating the normal reaction force to be equal to the weight. Parts (b) and (c) were done well, and many were successful in part (d), although some omitted a force from their calculations.

Q31.

This question was the least popular and successful on the paper. Many candidates showed a lack of understanding of the directions in which forces act, although there were fewer instances of including inappropriate forces than in the past. Part (a)(i) was moderately successful but with many sign errors. Part (a)(ii) was not done well, with few candidates able to write down an equation that contained all the relevant forces. Part (b) proved a further test of understanding, this time testing Newton's first Law, of which a significant number of candidates lacked knowledge.

Q32.

The candidates were more successful with this question than had been expected.

Part (a) was quite well answered. A reasonable number of candidates formed the two equations of motion correctly and then went on to obtain the required value. Some candidates did not resolve the weight of the particle on the slope. A small number of candidates did not form an equation of motion for each particle.

Those candidates who realised that the acceleration was zero did well in part (b). There were quite a few candidates who tried to use the acceleration from part (a) when tackling part (b).

Q33.

This question was done very well. The answer that was given did seem to have helped some candidates. In part (c), a few candidates stated that the tension was zero, even though they had done both parts (a) and (b) correctly.

Q34.

Parts (a) and (b) were generally done well. In part (b), the printed answer helped many candidates, but some could not put together a convincing argument. Part (c) was more challenging. Some candidates did resolve horizontally and apply the friction law, but then got into difficulties solving the resulting equation. Other candidates did not take both of the steps needed to produce an equation for T and made very little progress. There was also some evidence of candidates using $\sin 30^\circ$ instead of $\cos 30^\circ$ when resolving horizontally.

Q35.

The force diagram asked for in this question was drawn fairly well, but the usual errors of forces in the direction of motion and missing forces were present on some scripts. Parts (b) and (c) were done fairly well and it appeared that many 'candidates' had been helped by the printed answer in part (b). There were some candidates who did not realise that they should consider the tension in the rope when finding the normal reaction. Part (d) was found to be very much more demanding, with relatively few correct responses. Candidate typically omitted one term from their equation or failed to solve correctly for T .